Expt No. 1 **Dt:**

DETERMINATION OF REACTIONS OF SIMPLY SUPPORTED BEAM

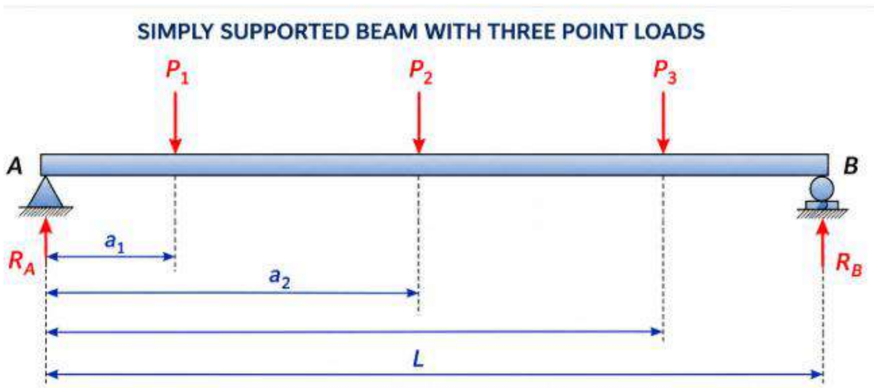
Aim: To write and execute python program for determination of reactions of simply supported beam

Interactive Development Environment (IDE) Used : Jupyter NoteBook

Programming Language Used : Python

Basic theory :.

A simply supported beam is subjected to three concentrated point loads acting at different locations along its span. Using the principles of static equilibrium, the reaction forces at the left and right supports are calculated from the applied loads and their distances from the left support.



Notations :

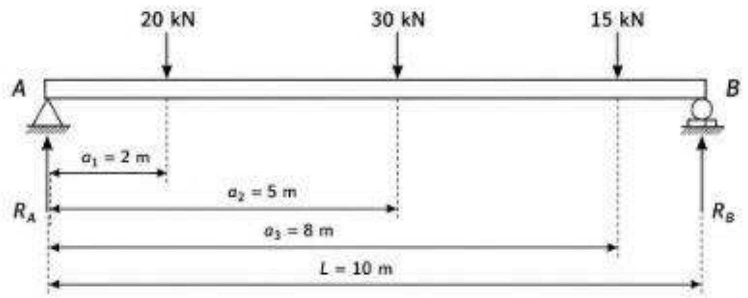
A	Left support (Pinned Support)
B	Right support (Roller Support)
$\mathbf{P_1}$	First downward point load acting on the beam (kN)
$\mathbf{P_2}$	Second downward point load acting on the beam (kN)
$\mathbf{P_3}$	Third downward point load acting on the beam (kN)
$\mathbf{a_1}$	Distance from the left support (A) to the first load $\mathbf{P_1}$ (m)
$\mathbf{a_2}$	Distance from the left support (A) to the first load $\mathbf{P_2}$ (m)
$\mathbf{a_3}$	Distance from the left support (A) to the first load $\mathbf{P_3}$ (m)
L	Total span (length) of the beam between supports A and B (m)
$\mathbf{R_A}$	Upward reaction force at the left support A (kN)
$\mathbf{R_B}$	Upward reaction force at the right support B (kN)

$$\sum M_A = 0$$

$$R_B = \frac{P_1 a_1 + P_2 a_2 + P_3 a_3}{L}$$

$$R_A = P_1 + P_2 + P_3 - R_B$$

Worked Example:



1. Calculate R_B

$$R_B = \frac{P_1 a_1 + P_2 a_2 + P_3 a_3}{L}$$

$$R_B = \frac{(20 \times 2) + (30 \times 5) + (15 \times 8)}{10}$$

$$R_B = \frac{40 + 150 + 120}{10}$$

$$R_B = \frac{310}{10}$$

$$\boxed{R_B = 31.00 \text{ kN (upward)}}$$

2. Calculate R_A

$$R_A = P_1 + P_2 + P_3 - R_B$$

$$R_A = 20 + 30 + 15 - 31.00$$

$$R_A = 65 - 31.00$$

$$\boxed{R_A = 34.00 \text{ kN (upward)}}$$

Python Concepts used:

Python Concept	Description	Example(s)
Comments (#)	Comments are used to explain the purpose of the program or specific statements. They improve the readability of the code and are ignored during program execution.	# Input beam span # Calculate reactions
input() Function	Used to accept input from the user during program execution. The value entered by the user is stored in a variable.	name = input("Enter your name: ") age = input("Enter your age: ")
Type Conversion (int(), float()**)**	Converts the input value from one data type to another. It is commonly used to convert string input into numeric values for calculations.	age = int(input("Enter age: ")) salary = float(input("Enter salary: "))
print() Function	Used to display text, variable values, and calculation results on the screen.	print("Welcome to Python") print("Sum =", a + b)

Python Program:

```
# Program to calculate reactions of a simply supported beam
# carrying three point loads
# Input beam span
L = float(input("Enter the span of the beam (m): "))
# Input point loads
P1 = float(input("Enter First Point Load P1 (kN): "))
P2 = float(input("Enter Second Point Load P2 (kN): "))
P3 = float(input("Enter Third Point Load P3 (kN): "))
# Input distances from left support A
a1 = float(input("Enter distance of P1 from Left Support A (m): "))
a2 = float(input("Enter distance of P2 from Left Support A (m): "))
a3 = float(input("Enter distance of P3 from Left Support A (m): "))
# Calculate reactions
RB = (P1 * a1 + P2 * a2 + P3 * a3) / L
RA = P1 + P2 + P3 - RB
# Display results
print("\n------ REACTIONS AT SUPPORTS ------")
print("Reaction at Left Support (RA) = ", round(RA, 2), "kN")
print("Reaction at Right Support (RB) = ", round(RB, 2), "kN")
```

Sample Input :

$L = 10m$, $P_1 = 20KN$, $P_2 = 30KN$, $P_3 = 15KN$, $a_1 = 2m$, $a_2 = 5m$, $a_3 = 8m$

Sample Output:

Reaction at Left Support (RA) = 34.00 kN

Reaction at Right Support (RB) = 31.00 kN

Result :

Python program was written and executed to determine the support reactions of simply supported beam.